Practice Set 16 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Outline the management of muscle invasive TCC. Discuss the role of adjuvant & neoadjuvant chemotherapy.

Type: LAQ (25 marks) (MIBC management also answered in Practice Set 12 Paper 2, Q2; neoadjuvant therapy in 2025-Winter Paper 4, Q1)

Introduction

- MIBC (muscle-invasive bladder cancer; TCC, transitional cell carcinoma, now called urothelial carcinoma) is cT2–T4a disease.
- Nearly half of patients treated by RC (radical cystectomy) alone develop metastases from occult micrometastases, so cure needs local plus systemic therapy, planned by an MDT (multidisciplinary team).

Staging

- Complete TURBT (transurethral resection of bladder tumour) with detrusor muscle; MRI (magnetic resonance imaging) with VI-RADS (Vesical Imaging-Reporting and Data System); CT (computed tomography) urogram and chest CT; bone scan if symptoms.
- Assess fitness and cisplatin eligibility.

Management algorithm

```
cT2–T4a N0 M0
 ├─ Cisplatin-eligible → cisplatin NAC (neoadjuvant chemotherapy)
 │      → RC + bilateral PLND (pelvic lymph node dissection)
 ├─ Cisplatin-ineligible → RC + PLND directly
 │      (pT3–4 or N+ after either → adjuvant nivolumab or chemotherapy)
 ├─ Selected, motivated → trimodality therapy (TMT)
 └─ Unfit → chemoradiation / palliative radiotherapy
cN+ → induction chemotherapy → consolidation RC in responders
M1  → systemic therapy
```

1. Radical cystectomy (standard of care)

- Men: cystoprostatectomy. Women: anterior exenteration; organ-sparing in selected cases.
- **Standard bilateral PLND:** obturator, external and internal iliac nodes. **SWOG S1011:** extended PLND gave no survival benefit.
- Do not delay beyond **12 weeks** from diagnosis (worse survival). Diversion: ileal conduit, orthotopic neobladder or continent pouch.

2. Bladder preservation: TMT

- Visibly complete TURBT + radiotherapy (64 Gy, or 55 Gy in 20 fractions) + radiosensitiser. **BC2001:** 5-FU (fluorouracil) + mitomycin C improved 2-year locoregional disease-free survival (67% vs 48%).
- Best candidates: cT2, solitary tumour, **no CIS (carcinoma in situ), no hydronephrosis**, good bladder function.
- 5-year cancer-specific survival 50–84%; salvage cystectomy 10–30%.
- **Also know:** partial cystectomy (solitary dome or urachal tumour, no CIS); chemotherapy alone or radical TURBT alone is not recommended.

3. Metastatic disease

- First line: **EV (enfortumab vedotin) + pembrolizumab** (EV-302: about doubled overall survival vs platinum). Alternatives: platinum chemotherapy then avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

Neoadjuvant chemotherapy (NAC)

Rationale

- Treats micrometastases early, while tumour burden is low, and is better tolerated before surgery.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

Eligibility: cT2–T4a N0 M0 and cisplatin-fit. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, grade ≥2 hearing loss or neuropathy, NYHA (New York Heart Association) class III–IV failure.

Trial	Regimen	Result
SWOG 8710	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC	pT0 38% vs 15%; median survival 77 vs 46 months; P = 0.06
BA06 30894 (EORTC/MRC)	CMV (cisplatin, methotrexate, vinblastine) ×3	16% relative reduction in mortality; 10-year OS (overall survival) 36% vs 30%
ABC meta-analysis (Advanced Bladder Cancer)	Cisplatin combinations	OS HR (hazard ratio) 0.86: **5% absolute 5-year survival gain**
VESPER	ddMVAC (dose-dense MVAC) ×6 vs GC (gemcitabine–cisplatin) ×4	Organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity
NIAGARA	Durvalumab + GC, then adjuvant durvalumab, vs GC	Better event-free survival (HR 0.68) and OS (HR 0.75)

- **Regimens:** ddMVAC 3–6 cycles or GC 4 cycles (EAU, AUA, NCCN: European Association of Urology, American Urological Association, National Comprehensive Cancer Network).
- **No benefit:** single-agent cisplatin, carboplatin regimens, neoadjuvant radiotherapy. Cisplatin-ineligible patients go straight to RC.
- The chapter calls neoadjuvant immunotherapy unproven; NIAGARA has since made perioperative durvalumab + GC an option.

Adjuvant therapy

	Neoadjuvant	Adjuvant
Selection	Clinical stage (understaging risk)	True pathological stage, but still overtreats
Delivery	Better tolerated	Post-RC morbidity; a quarter never receive it
Evidence	Level 1	Weaker: small, early-closed trials

Adjuvant chemotherapy

- Trials were small or closed early. **ABC 2022 update:** cisplatin-based adjuvant chemotherapy improves 5-year OS by about 6%. Benefit greatest in node-positive and extravesical disease.
- Use: cisplatin-naive patients with **pT3–4 or N+**: ddMVAC or GC.

Adjuvant immunotherapy

- **CheckMate 274:** nivolumab for 1 year after RC (pT3–4a or N+, with or without prior NAC): median disease-free survival **20.8 vs 10.8 months**.
- **IMvigor010** (atezolizumab): negative overall (included pT2); benefit in ctDNA (circulating tumour DNA)-positive patients.

- Use nivolumab if cisplatin-ineligible or declining chemotherapy, or with ypT3 or worse after NAC.
- **Also know:** adjuvant pelvic radiotherapy (Cairo trials) improves pelvic control in pT3–4; ctDNA-guided therapy is emerging.

Diagram to draw: bladder wall T-staging (T2a, T2b, T3, T4) beside the standard PLND template.

> **Key points to remember:**
> - NAC: cisplatin combinations only; ABC 5% absolute survival gain; SWOG 8710 pT0 38% vs 15%.
> - VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
> - CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
> - TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.

Q2. Desc. mgt optns for locally adv. CaP. Discuss complic. of b/l orchidectomy.

Type: LAQ (25 marks)

Assumed meaning: describe the management options for locally advanced carcinoma prostate, and discuss the complications of bilateral orchidectomy.

Definition and staging

- **Locally advanced CaP (carcinoma prostate):** T3–T4, N0 or N+, M0. Clinically: extracapsular spread (cT3a), seminal vesicle invasion (cT3b), adjacent organs (cT4). Today most "high-risk" disease is defined by grade and PSA (prostate-specific antigen) rather than palpation.
- At least 10% of new diagnoses; these men contribute a disproportionate share of cancer deaths.
- **Staging:** mpMRI (multiparametric magnetic resonance imaging) for local extent; bone scan + CT (computed tomography), or **PSMA PET (prostate-specific membrane antigen positron emission tomography)**, which is more accurate (proPSMA: 92% vs 65%).
- Untreated, Gleason 8–10 disease carries high cancer mortality, so fit men need treatment; no single modality suffices, hence **combined therapy**.

Management options

1. Radiotherapy + long-term ADT (androgen deprivation therapy): the commonest standard

- Dose-escalated EBRT (external-beam radiotherapy [RT], ≥78 Gy) with or without pelvic node RT; **brachytherapy boost** improves biochemical control (ASCENDE-RT).
- ADT works synergistically with RT: neoadjuvant, concurrent and adjuvant, **18–36 months** for high risk.

Trial	Finding
EORTC 22863	3 years of goserelin with RT: 5-year overall survival 79% vs 62%
RTOG 85-31 / 92-02	Long adjuvant ADT better; OS gain in Gleason 8–10
TROG 96.01	6 months better than 3 months
SPCG-7; MRC PR07	ADT + RT beats ADT alone (overall survival gain)
STAMPEDE (high-risk M0)	Adding 2 years of abiraterone: metastasis-free survival HR (hazard ratio) 0.53
RTOG 0521; GETUG 12	Docetaxel added modest benefit

2. RP (radical prostatectomy) + extended PLND (pelvic lymph node dissection)

- For fit men with **completely resectable** disease (cT3a, selected cT3b, limited nodes). Clinical overstaging occurs in 7–27%, so cT3 alone is no bar.
- **Why surgery:** exact pathological staging, removal of the primary (an untreated prostate causes retention, haematuria and ureteric obstruction that ADT does not prevent), and RT kept for salvage.
- Disease-specific survival 85–92% at 5 years and 79–82% at 10 years, but biochemical recurrence is common, so multimodal care is planned.
- **Neoadjuvant ADT before RP:** clinically downstages, but no survival or PSA benefit.
- **Adjuvant/salvage RT:** EORTC 22911, SWOG 8794, ARO 96-02 showed benefit; RADICALS-RT and RAVES favour **early salvage RT** (PSA rise) for most men.
- **pN+:** adjuvant ADT (ECOG 7887: better survival) ± RT.

3. ADT alone

- For men unfit for, or declining, local therapy. Early ADT delays progression; survival gain mainly in aggressive disease. In fit men it is inferior to ADT + RT.
- Options: bilateral orchidectomy, LHRH (luteinising hormone-releasing hormone) agonist or antagonist; bicalutamide 150 mg monotherapy preserves libido.

4. Others

- Watchful waiting with deferred ADT in elderly comorbid men.
- **Also know:** cryotherapy and HIFU (high-intensity focused ultrasound) remain investigational; TURP (transurethral resection of the prostate) or stents for local obstruction; trial enrolment.

Bilateral orchidectomy

- Removes about 95% of testosterone: **castrate level (<20 ng/dL) within 12 hours**. Cheap, no compliance issues, no flare, so useful in impending cord compression. Simple or subcapsular (all Leydig tissue must be removed).
- Versus LHRH agonists: fewer cardiac events, fractures and peripheral arterial disease reported.

A. Surgical complications

- Scrotal haematoma (commonest), wound infection, chronic scrotal or groin pain.
- Psychological: empty scrotum and altered self-image. Prevent with subcapsular or epididymis-sparing technique, or a testicular prosthesis.
- Irreversible: intermittent ADT is impossible.

B. Systemic effects of castration

Effect	Details and management
Hot flushes	Commonest (50–80%); loss of hypothalamic oestrogen signalling; venlafaxine, gabapentin, megestrol or cyproterone if bothersome
Sexual dysfunction	Loss of libido, ED (erectile dysfunction), penile shortening; ED aids help selected men
Osteoporosis, fractures	Fracture risk up 20–45%; baseline DEXA (dual-energy X-ray absorptiometry), FRAX; calcium, vitamin D, exercise; **denosumab 60 mg 6-monthly** or zoledronic acid
Metabolic	Weight gain, fat gain, **sarcopenia**, insulin resistance, diabetes (HR about 1.34), metabolic syndrome
Cardiovascular	Coronary events, sudden cardiac death, thromboembolism
Anaemia	Normocytic; reversible after stopping
Others	Gynaecomastia (≤5%), fatigue, depression, cognitive change

- **Also know:** risk of castration resistance over time; check testosterone with rising PSA.

Diagram to draw: T3a, T3b and T4 extension of a prostate tumour, and subcapsular orchidectomy.

> **Key points to remember:**
> - Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
> - RT + 18–36 months ADT: EORTC 22863 overall survival 79% vs 62%; STAMPEDE abiraterone metastasis-free survival HR 0.53.
> - Neoadjuvant ADT before RP: no survival benefit; early salvage RT favoured.
> - Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

Q3a. Vesicourodynamics

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 4, Q3d)

Assumed meaning: videourodynamics (VUDS), i.e. urodynamics with simultaneous fluoroscopic imaging of the bladder and urethra.

Definition

- **VUDS:** multichannel urodynamics (pressure–flow ± EMG [electromyography]) recorded with fluoroscopy, using **contrast as the filling fluid**.

Set-up

- Vesical and rectal catheters (Pdet [detrusor pressure] = Pves − Pabd), flowmeter, EMG; C-arm or tilting table.
- SUI (stress urinary incontinence) views standing; void in the usual position. **Screen only at key moments** (provocation, pressure rise, voiding).

What fluoroscopy adds

Situation	VUDS finding
NLUTD (neurogenic lower urinary tract dysfunction)	**Gold standard**; DESD (detrusor–external sphincter dyssynergia) and **DISD** (detrusor–internal sphincter dyssynergia, seen only on VUDS); trabeculation, diverticula
VUR (vesicoureteric reflux)	Pressure and volume at onset, grade; reflux "pops off" pressure, so compliance looks falsely safe
Bladder neck	Only way to assess it; **PBNO** (primary bladder neck obstruction)
Female obstruction	**Nitti criteria:** radiographic narrowing between bladder neck and meatus with a sustained detrusor contraction
Dysfunctional voiding	"Spinning-top" urethra at the external sphincter
SUI	Leakage with ALPP (abdominal leak point pressure); open bladder neck at rest suggests ISD (intrinsic sphincter deficiency)
Male obstruction	Localises prostatic obstruction, bladder neck contracture or stricture

PBNO in young men: type I high pressure, low flow; type II normal pressure, low flow, narrow bladder neck; type III delayed bladder-neck opening.

Indications

- Known or suspected NLUTD; unexplained retention in women; young men with LUTS (lower urinary tract symptoms); complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant; children with PUV (posterior urethral valves) or dysraphism.
- Shows risk factors: poor compliance, DLPP (detrusor leak point pressure) >40 cm H_2O, DESD.

Limitations

- Radiation, cost, space, contrast allergy. EAU (European Association of Urology): optimum test in NLUTD; AUA (American Urological Association): lowest dose that answers the question.

Diagram to draw: VUDS trace (Pves, Pabd, Pdet, flow, EMG) beside a voiding cystourethrogram image.

> **Key points to remember:**
> - VUDS = urodynamics + fluoroscopy with contrast.
> - Only test for bladder neck dysfunction, PBNO and DISD.
> - Reflux pop-off makes compliance look falsely good.

Q3b. Early surgical intervention in urethral injuries

Type: SAQ (10 marks) (posterior urethral injury also answered in Practice Set 5 Paper 3, Q1)

Principles

- Key decision: **immediate repair versus deliberate delay**. Clean, localised injuries are repaired at once; PFUI (pelvic fracture urethral injury) is drained and reconstructed later.
- **Blood at the meatus mandates RUG (retrograde urethrogram)** before catheterisation. Grades: posterior partial, minor or major distraction; anterior contusion, incomplete or complete.

Posterior urethra (PFUI)

- **SPC (suprapubic cystostomy) is the standard first step:** percutaneous 16 Fr Foley via peel-away sheath, or open. Safe with pelvic plating.
- **Primary realignment** (AUA [American Urological Association] Urotrauma option): brief attempt in a stable patient, acutely or within days, by combined antegrade and retrograde flexible cystoscopy (avoid prolonged attempts).
  - Catheter out at 4–6 weeks; keep the SPC; remove it 7–14 days later if voiding well.
  - Most still form a **1–2 cm stenosis**; **no difference** in urethroplasty rates vs SPC. Realignment does not add ED (erectile dysfunction) or incontinence.
- Partial tear: one gentle catheter attempt.
- **Immediate open repair** only for **bladder neck or rectal injury**, or when laparotomy is needed anyway.
- **Also know:** female urethral injuries are usually repaired early; in boys tears may extend into the bladder neck.

Anterior urethra

Injury	Early management
Contusion, incomplete straddle tear	Urethral catheter alone
Major straddle injury	SPC (standard); primary realignment for lesser injuries; delayed anastomotic urethroplasty
Low-velocity gunshot or stab	**Immediate primary repair:** debride minimally, spatulate, tension-free anastomosis over a catheter
High-velocity or close-range shotgun	SPC, staged repair
Penile fracture with urethral tear	**Immediate repair:** partial tear oversewn; complete tear debrided and anastomosed; catheter 2–3 weeks

- Penile fracture: urethral injury in about 20% of Western and 3% of Asian series; **intraoperative flexible cystoscopy** before catheterisation. Limit spongiosal debridement.

Diagram to draw: RUG showing partial vs complete posterior urethral disruption.

> **Key points to remember:**
> - Blood at meatus → RUG; SPC is standard for PFUI.
> - Primary realignment: brief, endoscopic; equal urethroplasty rates.
> - Immediate repair: penile fracture urethra, low-velocity penetrating injury, bladder neck/rectal injury.

Q3c. TVT

Type: SAQ (10 marks) (TVT versus TOT also answered in 2021-Summer Paper 2, Q3e)

Definition and history

- **TVT (tension-free vaginal tape):** retropubic polypropylene **MUS (mid-urethral sling)** for SUI (stress urinary incontinence); Ulmsten, 1995–1998.

Mechanism

- **Integral theory** (Petros and Ulmsten, 1990): continence needs pubourethral ligaments, the suburethral vaginal hammock and pubococcygeus. Hypermobility is a sign of their damage, not the cause.
- The loose tape secures the **posterior urethral wall** during stress (compression, dynamic kinking); it is not tight at rest.

Material

- **Type 1 (Amid) macroporous monofilament polypropylene**, pores >75 μm: allows macrophage entry and fibrous ingrowth, so it resists infection.

Indications

- Index SUI with urethral hypermobility; preferred over TOT (transobturator tape) in **ISD (intrinsic sphincter deficiency)** or recurrent SUI.
- Counsel about mesh risks and alternatives (colposuspension, autologous sling, bulking).

Technique

1. Local, spinal or general anaesthesia; lithotomy.
2. Two suprapubic stab incisions 2 cm lateral to midline; 1.5 cm midline vaginal incision 1.5 cm from the meatus; minimal dissection, endopelvic fascia not opened.
3. **Bottom-up** 5 mm trocar hugging the back of the pubis, through the retropubic space to the skin (catheter guide deflects the bladder).
4. **Cystoscopy with a 70° lens, full bladder**, after each pass; if perforated, re-pass more laterally.
5. Tension **loosely** with a clamp between tape and urethra; remove sheath.

Outcomes

- Ulmsten (1998): 91% cured at 1 year. **Ward–Hilton:** equal to colposuspension at 5 years. **TOMUS:** 80.8% (retropubic) vs 77.7% (TOT) at 12 months. Long-term cure varies 48–97% by definition.

Complications

Complication	Notes
Bladder perforation	4.5% (TOT 0.6%); re-pass and continue
Urethral injury	Abandon mesh; staged repair
Bleeding/haematoma	2–3%; rarely bowel or vascular injury
Voiding dysfunction	Surgery needed 2.7%; incise tape in midline within 3 months
Mesh exposure	About 2%
Others	De novo urgency, suprapubic pain (groin pain less than TOT)

Diagram to draw: coronal pelvis with U-shaped retropubic tape at mid-urethra.

Key points to remember:

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Cystoscopy after each trocar pass; bladder perforation 4.5%.

Q3d. Undescended testis

Type: SAQ (10 marks) (also answered in 2018-Summer Paper 3, Q3a and 2024-Winter Paper 1, Q2)

Definition and classification

- Testis not in the normal scrotal position: **palpable** (canalicular, high scrotal, **ectopic**, commonest at the superficial inguinal pouch) or **non-palpable** (intra-abdominal, "peeping", vanished/absent).
- **Retractile:** brought down and stays; review yearly. **Acquired:** previously scrotal, ascends later.

Epidemiology and aetiology

- 2–4% of term boys at birth, 1–2% at 3 months; commoner in preterm infants.
- Descent needs Leydig hormones: **INSL3** (transabdominal phase) and **androgens** (inguinoscrotal phase), with genitofemoral nerve/CGRP (calcitonin gene-related peptide) guiding the gubernaculum.

Evaluation

- Warm room, frog-leg position; "milk" from the anterior superior iliac spine. Contralateral hypertrophy suggests vanished testis.
- **No ultrasound or MRI** (magnetic resonance imaging): they do not change management.
- **Bilateral non-palpable, or with hypospadias:** urgent karyotype and hormones to exclude **CAH** (congenital adrenal hyperplasia) or other DSD (differences of sex development).

Management

- **Observe to 6 months corrected age; operate after 6 months, preferably by 12, and by 18 months at the latest** (AUA [American Urological Association] 2014). Early surgery restores testicular growth.
- **Hormones (hCG [human chorionic gonadotrophin], LHRH [luteinising hormone-releasing hormone]): not recommended**; descent in about 20% only.

Situation	Operation
Palpable	Inguinal orchidopexy (mobilise cord, ligate processus, subdartos pouch) or scrotal (Bianchi)
Non-palpable	Examination under anaesthesia, then laparoscopy
Testis seen, vessels long	Laparoscopic orchidopexy (primary most successful)
Short vessels	**Fowler–Stephens** one- or two-stage (testicular vessels divided; survives on vasal collaterals) or Shehata traction
Vessels and vas enter ring	Inguinal or scrotal exploration; excise nubbin
Blind-ending vessels	Stop: vanished testis
Postpubertal, normal contralateral testis	Orchidectomy

- **Complications:** atrophy (higher after Fowler–Stephens), re-ascent, vas injury.

Prognosis

- **Fertility:** paternity about 80% unilateral, one-third bilateral.

- **Cancer:** 2–5-fold risk; orchidopexy after 10–11 years carries relative risk 5.8 (Walsh). Teach self-examination.

Diagram to draw: laparoscopic view of the internal ring: testis, vessels entering ring, blind-ending vessels.

> **Key points to remember:**
>
> - Operate from 6 months, preferably by 12, latest 18 months.
> - No imaging, no hormones; laparoscopy for non-palpable testis.
> - Bilateral non-palpable: exclude CAH/DSD.

Q3e. Sperm Retrival techniques

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 3, Q3b and 2021-Summer Paper 4, Q2)

Assumed meaning: sperm retrieval techniques.

Indications

- **OA (obstructive azoospermia):** CBAVD (congenital bilateral absence of the vas deferens), unreconstructable obstruction, failed or declined vasectomy reversal, back-up at reconstruction.
- **NOA (non-obstructive azoospermia):** only testicular retrieval works.
- Anejaculation: vibratory stimulation or electroejaculation first.
- Retrieved sperm need **ICSI (intracytoplasmic sperm injection)**; cryopreserve spares.

Techniques

Technique	Method	Best for	Drawback
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput, local anaesthesia	OA	Few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened at 16–25×; micropipette by **capillary action**, never suction	**OA (best)**	Microsurgical skill
TESA (testicular sperm aspiration)	23 G needle, suction, many passes	OA	Immotile sperm, small yield; poor in NOA
Percutaneous core biopsy	14 G biopsy gun	OA	Blind; vessel injury
TESE (testicular sperm extraction)	Open biopsy, one or more sites	OA, NOA	Blind sampling
Micro-TESE	Testis bivalved; larger opaque tubules picked under the microscope	**NOA (best)**	Long operation

- **MESA:** move proximally until motile sperm appear (most motile in the efferent ductules); 4–12 pipettes, frozen in aliquots.
- **Micro-TESE (Schlegel 1999):** spermatogenesis in NOA is **focal**. Plump tubules hold sperm; thin white ones are Sertoli cell–only. Map positive sites; if negative, explore the other testis.

Results

- OA: retrieval almost always succeeds. NOA micro-TESE: sperm in **about half**.

Condition	Retrieval
Post-orchidopexy	74%
AZFc (azoospermia factor c) deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
Sertoli cell–only	30%
AZFa/AZFb deletion	**0%: do not operate**

- NOA pregnancies: more miscarriage (aneuploidy, DNA damage). Frozen sperm equal fresh, except in spinal cord injury.

Complications

- Haematoma (commonest), infection, epididymal obstruction (PESA), testicular atrophy or fall in testosterone (TESE).
- **Also know:** post-mortem retrieval needs prior consent; microfluidic sperm sorting.

Diagram to draw: micro-TESE: bivalved testis with plump tubules beside thin Sertoli cell–only tubules.

> **Key points to remember:**
>
> - OA: MESA (capillary action); NOA: micro-TESE.
> - Micro-TESE finds sperm in about half of NOA.
> - AZFa/AZFb deletion: 0%, do not operate.

Q3f. Salvage procedures for Complicated PUJO.

Type: SAQ (10 marks)

Assumed meaning: salvage options for recurrent PUJO (pelviureteric junction obstruction) after a failed pyeloplasty or endopyelotomy.

Definition and causes

- **Failure:** persistent pain or infection, unchanged or worse hydronephrosis, obstructed renogram or falling function. Most failures occur within 2 years, but up to 30% appear later.
- **Causes:** missed crossing vessel, ischaemic or tense anastomosis, urine leak with fibrosis, kinking, non-dependent anastomosis, infection.

Evaluation

- Diuretic renogram (MAG3, mercaptoacetyltriglycine): split function and drainage.
- CT (computed tomography) angiography for a crossing vessel; retrograde or antegrade pyelogram for stricture length.
- Drain first (stent or nephrostomy) if infected or function is uncertain, then reassess.

Salvage options

Option	Indication	Results/notes
Endopyelotomy (antegrade or retrograde ureteroscopic, laser)	Short stricture, no significant crossing vessel, moderate hydronephrosis	About 70% after failed pyeloplasty; lateral incision; stent 4–6 weeks
Redo pyeloplasty (open, laparoscopic, robotic)	Most failures; failed endopyelotomy	About 83–89%; wide mobilisation of kidney and ureter; transperitoneal (scarred retroperitoneum)
Ureterocalicostomy	Scarred or small intrarenal pelvis, dilated lower-pole calyx	About 70% in redo series; **resect** lower-pole parenchyma (not nephrotomy)
Buccal mucosa onlay graft	Long scarred PUJ or proximal ureter, dense scar	Robotic with omental wrap; failure 15–25% (CORRUS)
Appendiceal onlay flap	Right-sided proximal stricture	Pedicled on appendiceal artery
Flaps (spiral, vertical), downward nephropexy	Gain length	Combine with redo repair
Ileal ureter	Long defect	Contraindicated if creatinine >2 mg/dL, bladder dysfunction, IBD (inflammatory bowel disease), radiation
Autotransplantation	Solitary or poor contralateral kidney; other repairs impossible	About 90% long term
Nephrectomy	Function <15% with normal contralateral kidney	Laparoscopic

- Rule of thumb: failed pyeloplasty → try endourology first if anatomy suits; failed endopyelotomy → redo pyeloplasty.
- **Also know:** robotic near-infrared fluorescence (indocyanine green) to identify ureter and perfusion; Davis intubated ureterotomy (historical).

Diagram to draw: ureterocalicostomy: lower-pole parenchyma excised, spatulated ureter sewn to calyx.

Key points to remember:

- Redo pyeloplasty is most successful (about 83–89%); endopyelotomy about 70%.
- Ureterocalicostomy: resect lower-pole parenchyma; about 70% in redo series.
- Function <15% with normal other kidney: nephrectomy.

Rapid Revision Sheet – Practice Set 16 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.

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- RT + 18–36 months ADT: EORTC 22863 overall survival 79% vs 62%; STAMPEDE abiraterone metastasis-free survival HR 0.53.
- Neoadjuvant ADT before RP: no survival benefit; early salvage RT favoured.
- Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

Q3a. Vesicourodynamics (SAQ)

- VUDS = urodynamics + fluoroscopy with contrast.
- Only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

Q3b. Early surgical intervention in urethral injuries (SAQ)

- Blood at meatus → RUG; SPC is standard for PFUI.
- Primary realignment: brief, endoscopic; equal urethroplasty rates.
- Immediate repair: penile fracture urethra, low-velocity penetrating injury, bladder neck/rectal injury.

Q3c. TVT (SAQ)

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Cystoscopy after each trocar pass; bladder perforation 4.5%.

Q3d. Undescended testis (SAQ)

- Operate from 6 months, preferably by 12, latest 18 months.
- No imaging, no hormones; laparoscopy for non-palpable testis.
- Bilateral non-palpable: exclude CAH/DSD.

Q3e. Sperm Retrival techniques (SAQ)

- OA: MESA (capillary action); NOA: micro-TESE.
- Micro-TESE finds sperm in about half of NOA.
- AZFa/AZFb deletion: 0%, do not operate.

Q3f. Salvage procedures for Complicated PUJO. (SAQ)

- Redo pyeloplasty is most successful (about 83–89%); endopyelotomy about 70%.
- Ureterocalicostomy: resect lower-pole parenchyma; about 70% in redo series.
- Function <15% with normal other kidney: nephrectomy.